

SymPy Bug Card

Bug 36: Wrong-Sign Antiderivative on the Negative Real Branch

Task. Compute an indefinite integral:

$$\int \frac{dx}{x\sqrt{x^2+1}}.$$

SymPy's answer.

$$\text{integrate}\left(\frac{1}{x\sqrt{x^2+1}}, x\right) = -\operatorname{asinh}\left(\frac{1}{x}\right).$$

Correct answer. The answer must be branch-correct; on the negative real branch an antiderivative is

$$\operatorname{asinh}\left(\frac{1}{x}\right).$$

Explanation. Differentiating SymPy's result gives $1/(x^2\sqrt{1+x^{-2}})$. At $x = -2$, this is $1/(2\sqrt{5})$, but the requested integrand is $-1/(2\sqrt{5})$. The mismatch comes from losing the sign in $\sqrt{x^2} = |x|$, so the returned expression is valid only on one real branch.

Likely root cause. In `sympy/integrals/meijerint.py`, the indefinite Meijer path effectively substitutes through x^2 while an odd $1/x$ factor is present, then hyperexpands to an unconditional one-branch antiderivative. The deduplication assessment classifies this as a new instance of a known failure mode.

Artifact (GitHub).

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